Aryan Vadrevu

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SUMMARY

Mechanical Engineering student with hands-on expertise in SolidWorks, ANSYS, MATLAB, and FEA, seeking entry-level mechanical engineering internships. Experienced in mechanical design, structural analysis, and system-level modeling through Formula SAE Powertrain and multiple engineering projects, with project leadership experience directing a team from concept through fabrication and delivery. Certified in SolidWorks and manufacturing machining operations, and currently pursuing FE (EIT) certification. Passionate about automotive engineering, robotics, and autonomous systems, with a strong foundation in applying engineering principles to real-world design and manufacturing challenges.

EDUCATION

Rutgers University, New Brunswick, NJ
Bachelors in Mechanical Engineering
Expected Graduation : May 2027
Minor : Mathematics (Completed in Dec 2025)

TECHNICAL SKILLS

CAD / Simulation: SolidWorks (CSWP), ANSYS (FEA, Static Structural, Thermal), SolidWorks Motion and Simulation
Programming: MATLAB, Python, Arduino (C++)
Manufacturing: Manual Mill, Lathe, 3D Printing / Additive Manufacturing, Woodworking, Machining & Tolerancing
Software: MS Office (Word, Excel, PowerPoint, Access)

EXTRACURRICULAR ACTIVITIES

Accessible 4 All (A4A) – Project Lead

- Lead a team of approximately 15 members designing and building adaptive equipment — modified ride-on cars, tricycles, and custom toys — for children with disabilities.
- Work directly with families to gather requirements, define mobility and accessibility needs, and translate them into design specifications for each build.
- Own projects end to end, from concept and CAD modeling through fabrication, assembly, and final delivery of a finished, tested product to the family.
- Coordinate woodworking, machining, and CAD workstreams across the team, delegating tasks, reviewing designs, and keeping builds on schedule.

Rutgers Formula Racing (FSAE) - Powertrain Team Member

- Designed and modeled accumulator mounts in SolidWorks to meet FSAE compliance and packaging constraints.
- Assisted in the design and machining of motor shafts for the electric powertrain system.
- Conducted FEA in ANSYS to evaluate stress distribution, deformation, and factor of safety on critical components.
- Supported design validation through simulation-driven decision making, reducing prototype iteration cycles.

PROJECTS

Structural & Thermal Analysis — ANSYS FEA Study

- Performed static structural analysis on mechanical components to evaluate stress concentration, deformation, and factor of safety under realistic loading conditions.
- Conducted thermal analysis to simulate heat distribution and identify potential failure points in high-temperature assemblies.
- Validated simulation results against theoretical calculations to ensure accuracy and reliability of FEA models.
- Generated detailed reports with contour plots, mesh refinement studies, and design recommendations to optimize component geometry.

Economical Gimbal System

- Designed a multi-axis handheld camera gimbal from concept to final assembly in SolidWorks, iterating through initial sketches, Fusion 360, and final SolidWorks models to optimize geometry and manufacturability.

- Conducted material selection and analysis, evaluating ABS for 3D-printed structural components based on mechanical properties, weight, cost, and compatibility with additive manufacturing processes.
- Performed load and stress analysis including bending moment calculations, stress concentration evaluation, and center-of-mass studies to validate structural integrity under realistic loading conditions.
- Applied GD&T and tolerancing principles to critical mating features, including a third-axis pin (18mm diameter) and 0.25" bolt/nut fastener interfaces, ensuring proper fit and function across all assemblies.
- Ran FEA in SolidWorks Simulation to assess stress distribution and deformation under loading, and conducted motion studies to verify gimbal axis rotation and pitch kinematics.
- Optimized design for low-cost scalable manufacturing, achieving a per-unit material cost of ~$8.78 using bulk ABS filament and standard hardware.

Bicycle Design & FEA Project

- Designed and modeled a complete bicycle assembly in SolidWorks with focus on functionality, aesthetics, and mechanical integrity.
- Performed FEA on the frame to simulate stress distribution under static and dynamic loading conditions, informing material and geometry choices.
- Utilized SolidWorks Motion to simulate wheel rotation, steering, and pedaling kinematics for mechanism validation.
- Applied realistic materials and rendered the model for presentation and portfolio documentation.

Simplified Liquid Rocket Engine

- Collaborated with a team to design and assemble a simplified liquid rocket engine model focusing on propulsion fundamentals.
- Modeled combustion chamber, injector plate, and nozzle in SolidWorks, ensuring efficient mixing and high-pressure containment.
- Analyzed and optimized turbo pump and cooling mechanisms using fluid dynamics and structural analysis principles.
- Integrated multi-disciplinary engineering concepts (thermodynamics, combustion, fluid dynamics) into a cohesive system-level model.

Distance-Sensing Rotating Turret — Mechatronics Project

- Designed and built a semi-autonomous turret system integrating six mechatronic subsystems for object detection and projectile launching.
- Programmed an Arduino Mega to process sensor input, control actuators, and automate angle control and firing decisions in real time.
- Used MATLAB for numerical analysis and data visualization of sensor readings and system performance metrics.
- Achieved autonomous object detection and targeting, reinforcing practical skills in embedded systems and control logic

CERTIFICATES

Certified Solidworks Professional (CSWP) : Earned certification demonstrating advanced Solidworks expertise in complex assemblies, mates, interference detection and design optimization.
Certified Solidworks Associate (CSWA) : Earned certification demonstrating proficiency in SolidWorks modeling, assembly creation, and engineering drawing interpretation.
NCEES – Fundamentals of Engineering (FE) / Engineer in Training (EIT) – In Progress

ADDITIONAL SKILLS

- **Communication :** Strong verbal and written communication skills developed through presentations and team collaboration.
- **Teamwork :** Proven ability to work effectively in team environments.
- **Problem-Solving :** Analytical and critical thinking skills honed through coursework and practical engineering challenges.
- **Multitasking:** Proven ability to manage multiple projects simultaneously, ensuring timely completion while maintaining high-quality results.